Ratio-Sharp Global Defect Closure for Spherical Shells

Purely analytic support module

Abstract

This note proves the global strict proxy inequality for $0 < \rho \le 39/50$ from the first possible radial frequency onward. It combines the exact retained-remainder identity with the ratio-dependent slope cap, a tangent minorant of the shifted sawtooth primitive, a universal ball quarter-layer, and an exact scalar convexity argument. It contains no finite Bessel-zero staircase, event-state theorem, numerical root isolation, or computer-assisted premise.

1 The global defect argument

Let

$$A_{\rho,1} := \{x \in \mathbb{R}^3 : \rho < |x| < 1\}.$$

Fix $0 < \rho < 1$ and $K > 0$. For $a > 0$, define the zero-extended ball action

$$G_a(z) := \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \le z < a, \\ 0, & z \ge a, \end{cases} \tag{1}$$

and put

$$A(z) := G_K(z) - G_{\rho K}(z), \qquad T := A(0) = \frac{(1-\rho)K}{\pi}. \tag{2}$$

Write

$$[u]_< := \#\{n \in \mathbb{N} : n < u\}.$$

The phase reduction in the main paper supplies

$$N_D(A_{\rho,1}, K^2) \le P(\rho, K), \qquad P(\rho, K) := \sum_{\substack{\nu = \ell + 1/2 < K \\ \ell \in \mathbb{N}_0}} 2\nu\, [A(\nu) + \tfrac{1}{4}]_<. \tag{3}$$

Indeed, the channel phase is strictly smaller than $A(\nu) + 1/4$, so at an integral proxy wall the last radial integer remains excluded. The continuous payment is

$$W(\rho, K) := \int_0^K 2z A(z)\, dz = \frac{2(1-\rho^3)K^3}{9\pi}. \tag{4}$$

Action geometry and the exact defect

Differentiating (1) gives

$$q(z) := -A'(z) = \frac{1}{\pi}\begin{cases}\arccos\dfrac{z}{K} - \arccos\dfrac{z}{\rho K}, & 0 < z < \rho K,\\[4pt] \arccos\dfrac{z}{K}, & \rho K \le z < K.\end{cases} \tag{5}$$

Thus A decreases strictly from T to 0. Let $R : [0,T] \to [0,K]$ be its decreasing inverse and set

$$F(t) := R(t)^2, \qquad g(t) := -F'(t), \qquad \tau := A(\rho K) = KG_1(\rho). \tag{6}$$

Write throughout

$$\theta := \arccos\rho.$$

Lemma 1 (Shape and endpoint data)**.** *With this notation, one has*

$$0 \le q(z) \le \frac{\theta}{\pi}, \qquad q \textit{ increases on } (0,\rho K), \qquad q \textit{ decreases on } (\rho K, K), \tag{7}$$

and

$$g \textit{ decreases on } (0,\tau), \qquad g \textit{ increases on } (\tau, T). \tag{8}$$

Moreover,

$$F(\tau) = \rho^2K^2, \qquad g(\tau) = \frac{2\pi\rho K}{\theta}, \qquad g(T) = \frac{2\pi\rho K}{1-\rho}. \tag{9}$$

The two formulas for g agree at $t = \tau$, and $g(t) = O(t^{-1/3})$ as $t \downarrow 0$.

Proof. On the inner branch, direct differentiation of (5) shows that $q' > 0$; on the outer branch $q' < 0$. Both one-sided values at $z = \rho K$ equal θ/π, proving (7).

If $z = R(t) > \rho K$, then

$$g(t) = \frac{2\pi z}{\arccos(z/K)}.$$

The quotient on the right increases with z, while z decreases with t, so g decreases on $(0,\tau)$. On the inner branch put

$$\delta(z) := \arccos\frac{z}{K} - \arccos\frac{z}{\rho K}.$$

The identity

$$\frac{\delta(z)}{z} = \int_\rho^1 \frac{ds}{s\sqrt{s^2K^2 - z^2}}$$

shows that $\delta(z)/z$ increases with z. Hence $g(t) = 2\pi z/\delta(z)$ increases as t increases and z decreases. The endpoint values follow by substitution and the limit $z \downarrow 0$. Finally, writing $z = K\cos u$ on the outer branch gives $t \asymp u^3$ and $g \asymp u^{-1}$. □

Put

$$x_n := n - \frac{1}{4}, \qquad M(r) := \#\{\ell \in \mathbb{N}_0 : \ell + \tfrac{1}{2} < r\}, \qquad N := \#\{n \ge 1 : x_n < T\}. \tag{10}$$

For an active n, write $r_n := R(x_n)$ and $m_n := M(r_n)$.

Lemma 2 (Strict layer cake)**.** *For every* $0 < \rho < 1$ *and* $K > 0$*,*

$$P(\rho, K) = \sum_{x_n < T} m_n^2, \qquad W(\rho, K) = \int_0^T F(t)\, dt. \tag{11}$$

Both identities retain every radial and angular equality wall.

Proof. For a half-integer $\nu = \ell + 1/2$,

$$[A(\nu) + \tfrac{1}{4}]_< = \#\{n \geq 1 : x_n < A(\nu)\}.$$

Since A and R are decreasing, this is equivalent to $\nu < R(x_n)$. Interchanging the finite sums and using $\sum_{\ell=0}^{m-1}(2\ell + 1) = m^2$ proves the first identity. At $R(x_n) = m + 1/2$, strict angular counting gives $M(R(x_n)) = m$. The second identity is the area formula $\int_0^T R(t)^2 dt = \int_0^K 2zA(z)dz$. □

Define

$$\mathcal{E}_{\text{ang}} := \sum_{x_n < T} (m_n^2 - r_n^2), \qquad \mathcal{D}_{\text{rad}} := \int_0^T F(t)\, dt - \sum_{x_n < T} F(x_n). \tag{12}$$

Then

$$W - P = \mathcal{D}_{\text{rad}} - \mathcal{E}_{\text{ang}}. \tag{13}$$

The ratio-sharp angular payment

Proposition 3 (Ratio-sharp angular blocks)**.** *For* $N \geq 1$*,*

$$\mathcal{E}_{\text{ang}} < \frac{(1 - \rho^2)K^2}{6} - N d(\rho), \qquad d(\rho) := \frac{3\pi}{8 \arccos \rho} - \frac{1}{4}. \tag{14}$$

The inequality is strict at every common radial–angular wall. If $N = 0$*, then* $\mathcal{E}_{\text{ang}} = 0$*.*

Proof. Fix an active n and $t \in [n - 1, x_n]$. Since $R(t) \geq r_n$,

$$x_n - t = A(r_n) - A(R(t)) = \int_{r_n}^{R(t)} q(s)\, ds \leq \frac{\theta}{\pi}(R(t) - r_n).$$

Integration over the left block of length $3/4$ yields

$$r_n \leq \frac{4}{3} \int_{n-1}^{x_n} R(t)\, dt - \frac{3\pi}{8\theta}. \tag{15}$$

Strict angular counting gives $m_n < r_n + 1/2$, including when r_n is a half-integer, and hence

$$m_n^2 - r_n^2 < r_n + \frac{1}{4} \leq \frac{4}{3} \int_{n-1}^{x_n} R(t)\, dt - d(\rho). \tag{16}$$

The blocks are disjoint subsets of $[0, T]$. Therefore

$$\mathcal{E}_{\text{ang}} < \frac{4}{3} \int_0^T R(t)\, dt - N d(\rho).$$

The inverse area identity and scaling give

$$\int_0^T R(t)\, dt = \int_0^K A(z)\, dz = \frac{(1 - \rho^2)K^2}{8},$$

which proves (14). □

The tangent-envelope radial payment

Let

$$C_0(t) := \#\{n \ge 1 : x_n < t\}, \quad w(t) := t - C_0(t), \quad \mathfrak{W}(t) := \int_0^t w(s)\,ds, \quad \Psi(t) := \mathfrak{W}(t) - \frac{t}{4}. \quad (17)$$

On $0 \le t \le 3/4$,

$$\mathfrak{W}(t) = \frac{t^2}{2}, \qquad \Psi(t) = \frac{1}{2}\left(t - \frac{1}{4}\right)^2 - \frac{1}{32}. \quad (18)$$

On every later unit cell, direct integration gives

$$-\frac{1}{32} \le \Psi(t) \le \frac{3}{32}, \qquad \mathfrak{W}(t) \ge \frac{t}{4} - \frac{1}{32}. \quad (19)$$

Lemma 4 (Retained radial remainder)**.** *For all* $0 < \rho < 1$ *and* $K > 0$*,*

$$\mathcal{D}_{\mathrm{rad}} \ge \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau)\left(\frac{\tau}{4} + \frac{3}{32}\right) - \int_{(0,\tau)} \mathfrak{W}(t)\,dg(t). \quad (20)$$

Proof. Distributionally, $dw = dt - dC_0$. Stieltjes integration by parts gives $\mathcal{D}_{\mathrm{rad}} = \int_0^T w(t)g(t)dt$; the improper boundary term vanishes because $\mathfrak{W}(t)g(t) = O(t^{5/3})$ at zero. On the decreasing branch,

$$\int_0^\tau wg = \mathfrak{W}(\tau)g(\tau) - \int_{(0,\tau)} \mathfrak{W}\,dg.$$

On the increasing branch use $w = 1/4 + \Psi'$, $\int_\tau^T g = F(\tau)$, and integrate by parts. Since $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, the interface value cancels and

$$\mathcal{D}_{\mathrm{rad}} = \frac{F(\tau)}{4} + \frac{\tau g(\tau)}{4} + \Psi(T)g(T) - \int_{(\tau,T)} \Psi\,dg - \int_{(0,\tau)} \mathfrak{W}\,dg.$$

Because dg is positive on (τ, T),

$$\Psi(T)g(T) - \int_{(\tau,T)} \Psi\,dg \ge -\frac{g(T)}{8} + \frac{3g(\tau)}{32}.$$

This proves (20). Continuity of g and $\mathfrak{W}$ handles an interface on a sawtooth wall without an atom. □

Define the C^1 tangent envelope

$$L_\sharp(t) := \begin{cases} t^2/2, & 0 \le t \le 1/4, \\ t/4 - 1/32, & t \ge 1/4, \end{cases} \qquad L_\sharp'(t) = \min\{t, 1/4\}. \quad (21)$$

Equations (18)–(19) show

$$0 \le L_\sharp(t) \le \mathfrak{W}(t) \qquad (t \ge 0). \quad (22)$$

On $1/4 \le t \le 3/4$ the difference is $\frac{1}{2}(t - 1/4)^2$; after $3/4$ use (19).

Assume now

$$0 < \rho \le \frac{39}{50}, \qquad K \ge \frac{\pi}{1-\rho}. \quad (23)$$

The elementary turning estimate

$$G_1(s) = \frac{1}{\pi}\int_s^1 \arccos u\,du \geq \frac{2\sqrt{2}}{3\pi}(1-s)^{3/2} \tag{24}$$

implies

$$\tau \geq \frac{2\sqrt{2}}{3}\sqrt{1-\rho} \geq \frac{2\sqrt{2}}{3}\sqrt{\frac{11}{50}} > \frac{1}{4}.$$

The last inequality follows after squaring from $44/225 - 1/16 = 479/3600 > 0$.

On $(0,\tau)$, the measure $-dg$ is positive. Hence

$$\begin{aligned} -\int_{(0,\tau)} \mathfrak{W}\,dg &\geq -\int_{(0,\tau)} L_\sharp\,dg \\ &= -L_\sharp(\tau)g(\tau) + \int_0^\tau L_\sharp'(t)g(t)\,dt. \end{aligned} \tag{25}$$

There is no boundary contribution at zero. On the outer branch $t = A(z) = G_K(z)$ and $g(t)dt = -2z\,dz$, so

$$\int_0^\tau L_\sharp'(t)g(t)\,dt = \int_{\rho K}^K 2z\min\{G_K(z), 1/4\}\,dz. \tag{26}$$

Since $\tau > 1/4$, the interface factor becomes

$$g(\tau)\left(\frac{\tau}{4} + \frac{3}{32} - L_\sharp(\tau)\right) = \frac{g(\tau)}{8}.$$

Lemma 4 and (9) now give

$$\mathcal{D}_{\mathrm{rad}} \geq \mathcal{H}_\sharp(\rho, K), \tag{27}$$

where

$$\mathcal{H}_\sharp(\rho, K) := \frac{\rho^2K^2}{4} - \frac{\pi\rho K}{4(1-\rho)} + \frac{\pi\rho K}{4\arccos\rho} + \int_{\rho K}^K 2z\min\{G_K(z), 1/4\}\,dz. \tag{28}$$

The simplified interface payment is asserted only on (23). Below $\tau = 1/4$, its unsimplified factor is $g(\tau)(\tau/4 + 3/32 - \tau^2/2)$.

The universal ball quarter-layer

Let R_B be the decreasing inverse of the restriction of G_K to $[0, K]$. Since $G_K(\rho K) = \tau > 1/4$, truncated layer cake gives

$$\frac{\rho^2K^2}{4} + \int_{\rho K}^K 2z\min\{G_K(z), 1/4\}\,dz = B_0(K) := \int_0^{1/4} R_B(t)^2\,dt. \tag{29}$$

Therefore

$$\mathcal{H}_\sharp(\rho, K) = B_0(K) - \frac{\pi\rho K}{4(1-\rho)} + \frac{\pi\rho K}{4\arccos\rho}. \tag{30}$$

From (24) and $G_K(z) = KG_1(z/K)$,

$$R_B(t) \geq K - CK^{1/3}t^{2/3}, \qquad C := \left(\frac{3\pi}{2\sqrt{2}}\right)^{2/3} = \frac{(3\pi)^{2/3}}{2}. \tag{31}$$

The right side is nonnegative for $0 \le t \le 1/4$: it is enough that $K \ge C^{3/2}/4 = 3\pi/(8\sqrt{2})$, whereas $K \ge \pi/(1-\rho) > \pi$. Squaring and integrating gives

$$B_0(K) \ge \frac{K^2}{4} - \alpha K^{4/3} + \beta K^{2/3}, \tag{32}$$

where

$$\alpha := \frac{6C}{5\,4^{5/3}}, \qquad \beta := \frac{3C^2}{7\,4^{7/3}}. \tag{33}$$

Combining (13), (14), (27), and (32), and using $T \ge 1$, hence $N \ge 1$, gives

$$\begin{aligned} W - P > \frac{1+2\rho^2}{12}K^2 - \alpha K^{4/3} + \beta K^{2/3} - \frac{\pi\rho K}{4(1-\rho)} \\ + \frac{\pi\rho K}{4\arccos\rho} + d(\rho). \end{aligned} \tag{34}$$

Exact scalar closure

Lemma 5 (Elementary angle bound)**.** *For* $0 \le \rho < 1$,

$$\arccos\rho \le \frac{\pi}{2}\sqrt{1-\rho}. \tag{35}$$

Proof. Write $\rho = \cos\theta$, $0 \le \theta \le \pi/2$, and $y = \theta/2$. Since $\sin y/y$ decreases on $[0, \pi/4]$,

$$\frac{\sin y}{y} \ge \frac{2\sqrt{2}}{\pi}.$$

Using $1 - \rho = 2\sin^2 y$ gives (35). □

Put $e := 1 - \rho$. It suffices to prove positivity of

$$\begin{aligned} \underline{\mathcal{G}}(\rho, K) := \frac{1+2\rho^2}{12}K^2 - \alpha K^{4/3} + \beta K^{2/3} - \frac{\pi\rho K}{4e} \\ + \frac{\rho K}{2\sqrt{e}} + \frac{3}{4\sqrt{e}} - \frac{1}{4}. \end{aligned} \tag{36}$$

Define

$$\mathsf{A} := \alpha\pi^{4/3}, \quad \mathsf{B} := \beta\pi^{2/3}, \quad C_1 := \frac{4}{3}\alpha\pi^{1/3}, \quad C_2 := \frac{2}{3}\beta\pi^{-1/3}. \tag{37}$$

Positive-side cubing and $157/50 < \pi < 22/7$ give

$$\mathsf{A} < \frac{49}{40}, \qquad \frac{179}{1000} < \mathsf{B} < \frac{1}{5}, \qquad C_1 < \frac{13}{25}, \qquad C_2 < \frac{1}{25}. \tag{38}$$

For completeness,

$$\mathsf{A}^3 = \frac{3^5\pi^6}{5^3 2^{10}}, \quad \mathsf{B}^3 = \frac{3^7\pi^6}{7^3 2^{20}}, \quad C_1^3 = \frac{3^2\pi^3}{5^3 2^4}, \quad C_2^3 = \frac{3^4\pi^3}{7^3 2^{17}}. \tag{39}$$

Thus each comparison in (38) is an exact rational cross multiplication after cubing.

Set

$$K_0 := \frac{\pi}{e}, \qquad x := e^{-1/6} \ge 1. \tag{40}$$

Direct substitution gives

$$\underline{\mathcal{G}}(\rho, K_0) = F_0(x), \tag{41}$$

where

$$\begin{aligned} F_0(x) := {} & \frac{\pi}{2}x^9 - \mathsf{A}x^8 - \frac{\pi^2}{12}x^6 + \mathsf{B}x^4 \\ & + \left(\frac{3}{4} - \frac{\pi}{2}\right)x^3 + \frac{\pi^2}{6} - \frac{1}{4}. \end{aligned} \tag{42}$$

The constant bounds imply

$$F_0(1) > \frac{8269}{30000} > \frac{1}{4}, \quad F_0'(1) > -\frac{14437}{6125} > -\frac{5}{2}, \quad F_0''(1) > \frac{18162}{1225} > 14. \tag{43}$$

For $x \geq 1$, group the leading terms as $x^5(252\pi x - 336\mathsf{A})$ and use $x^5 \geq x^3$ after the resulting coefficient is positive. Dropping the positive $24\mathsf{B}x$ term then gives

$$\begin{aligned} F_0'''(x) &> x^3\left(252\frac{157}{50} - 336\frac{49}{40} - 10\left(\frac{22}{7}\right)^2\right) - \frac{69}{14} \\ &\geq \frac{676141}{2450} > 0. \end{aligned} \tag{44}$$

Hence $F_0''(x) > 14$. With $y = x - 1 \geq 0$, Taylor's formula yields

$$F_0(x) > \frac{1}{4} - \frac{5}{2}y + 7y^2 = 7\left(y - \frac{5}{28}\right)^2 + \frac{3}{112} > 0. \tag{45}$$

A direct differentiation gives

$$\partial_K \underline{\mathcal{G}}(\rho, K_0) = D_0(x), \tag{46}$$

where

$$D_0(x) := \frac{\pi}{12}(3x^6 - 5 + 4x^{-6}) + \frac{1}{2}(x^3 - x^{-3}) - C_1x^2 + C_2x^{-2}. \tag{47}$$

At $x = 1$,

$$D_0(1) > \frac{1}{300}, \qquad D_0'(1) > \frac{54}{175}. \tag{48}$$

Moreover,

$$\begin{aligned} D_0''(x) &= \frac{\pi}{12}(90x^4 + 168x^{-8}) + 3x - 6x^{-5} - 2C_1 + 6C_2x^{-4} \\ &> \frac{15}{2}\frac{157}{50} - 3 - \frac{26}{25} = \frac{1951}{100} > 0. \end{aligned} \tag{49}$$

Thus $D_0(x) > 0$ for every $x \geq 1$.

Finally,

$$\partial_K^2 \underline{\mathcal{G}} = \frac{1 + 2\rho^2}{6} - \frac{4\alpha}{9}K^{-2/3} - \frac{2\beta}{9}K^{-4/3}. \tag{50}$$

For $K \geq K_0 \geq \pi$,

$$\alpha K^{-2/3} \leq \frac{\mathsf{A}}{\pi^2}, \qquad \beta K^{-4/3} \leq \frac{\mathsf{B}}{\pi^2}.$$

Consequently,

$$\partial_K^2 \underline{\mathcal{G}} > \frac{1}{6} - \frac{4(49/40) + 2(1/5)}{9(157/50)^2} = \frac{47447}{443682} > 0. \tag{51}$$

The positive base derivative remains positive, and

$$\underline{\mathcal{G}}(\rho, K) > 0 \qquad (K \geq K_0). \tag{52}$$

Theorem 6 (Global low- and middle-ratio proxy theorem)**.** *For*

$$0 < \rho \le \frac{39}{50}, \qquad K \ge \frac{\pi}{1-\rho},$$

one has

$$P(\rho, K) < W(\rho, K). \tag{53}$$

Consequently, $N_D(A_{\rho,1}, K^2) < W(\rho, K)$.

Proof. Equation (34), the angle bound, and (52) give

$$W - P > \underline{\mathcal{G}}(\rho, K) > 0.$$

The layer-cake and angular-block proofs retain every strict wall. □